

Appendices: Cubic Galileon Gravity on Nonlinear Scales with Hi-COLA for LSST Year 1

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References

- [1] Bill S. Wright, Ashim Sen Gupta, Tessa Baker, Georgios Valogiannis, and Bartolomeo Fiorini. Hi-cola: fast, approximate simulations of structure formation in horndeski gravity. *Journal of Cosmology and Astroparticle Physics*, 2023(03):040, March 2023. [1](#), [2](#)

A Sensitivity of the angular power spectra to f_ϕ

Here, we aim to get an idea of the effect of the modified gravity parameter f_ϕ on the angular power spectra we use for the forecasts in Section 6.2.2, going beyond the validation of the emulated boost in Section 6.1. In Figure 1, we consider the 3×2 pt residuals with respect to the corresponding GR angular power spectra:

$$\frac{C_{\text{CuGal}, AB}^{ij}(\ell) - C_{\text{GR}, AB}^{ij}(\ell)}{C_{\text{GR}, AB}^{ij}(\ell)},$$

where all quantities are defined in Section 5. To assess how this is affected by the HiCOLA error, we additionally plot the residual predicted from the EFT-RAMSES simulation (dotted, purple). We therefore estimated the ‘CuGal’ 3×2 pt datavector either by using the emulated boost or the boosts predicted from the EFT-RAMSES simulations.

Variations in f_ϕ produce deviations in the spectra that exceed the 1σ uncertainty level at several multipoles ℓ , for f_ϕ significantly smaller than 1. These regions therefore represent scales where modified gravity signatures of this Cubic Galileon model that are as of yet unconstrained may be observationally detectable for LSST Y1. However, the interpretation of these deviations is complicated by parameter degeneracies, including correlations with both cosmological and systematic nuisance parameters. We therefore refer to Section 6.2.2, for a more significant characterization of how our emulator can support potential detection of signatures of Cubic Galileon gravity.

A.1 Emulator Accuracy and Validation: comparison with N-body simulations

The Hi-COLA boost and the EFT-RAMSES boost (which, for consistency with the emulator, we have smoothed with a Savitzky–Golay filter) agree to sub-percent accuracy up to $k = 0.48h \text{ Mpc}^{-1}$ at $z = 0$ (compared to $k \leq 0.9h \text{ Mpc}^{-1}$ sub-percent agreement between Hi-COLA and the emulator), degrading to $\approx 5\%$ by $k \approx 1.2 h \text{ Mpc}^{-1}$ (compared to $\approx 1.8\%$ for the emulator). We can see the effect of propagating this error to the observables in the 3×2 pt angular power spectra in Figure 2, for the survey specifications described in Section 6.2.2.

We note that the error between Hi-COLA and EFT-RAMSES differs slightly from the Hi-COLA vs. ECOSMOG error reported in [1]. This difference is driven primarily by updates in

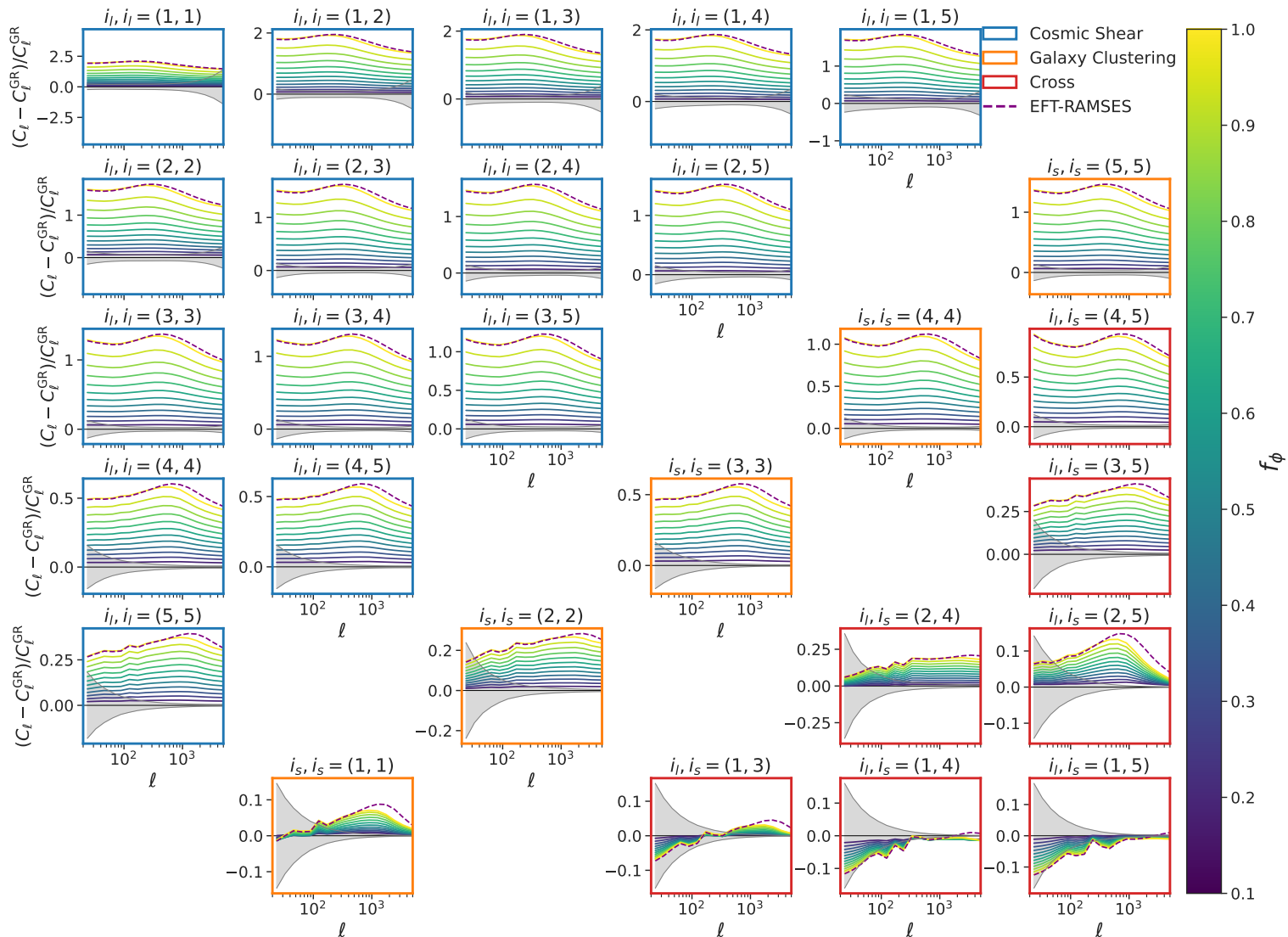


Figure 1: Sensitivity analysis of the angular power spectra to f_ϕ . We display the residual relative to the Λ CDM expectation, $(C_l^{cG}(\ell) - C^{GR}(\ell))/C^{GR}(\ell)$. The shaded area depicts the diagonal errors in the C_ℓ values using the DESC SRD LSST Y1 covariance: $\epsilon(\ell_i)/C^{GR}(\ell_i) = \sqrt{Cov_{ii}}/C^{GR}(\ell_i)$. In dotted purple, we also display the residual of the prediction of the angular power spectra based on the EFT-RAMSES simulation as described in section A.1.

the choice of cosmological parameters, and an increase in the resolution of and refinements to the force computations in EFT-RAMSES. In particular, the improvement in the computation of forces in this updated data are expected to drive an increase in the Hi-COLA error on small scales, and may necessitate recalibrating the k_{filter} parameter (see Appendix B.2 of [1]) against data from EFT-RAMSES.

We show in Section 6.2.3 that, for scale cuts (described in Section 5) that restrict the analysis to conservative baryonic cuts for weak lensing, and $k_{\text{max}} = 0.2 h \text{ Mpc}^{-1}$ for galaxy clustering and the cross-correlations, the COLA-induced error lies safely below the scales

used for this more conservative constraint for cosmological inference. This is also why a recalibration of the Hi-COLA k_{filter} parameter was not performed within this work, as the improved accuracy of Hi-COLA from performing this recalibration is expected to manifest on scales beyond the scale cuts considered.

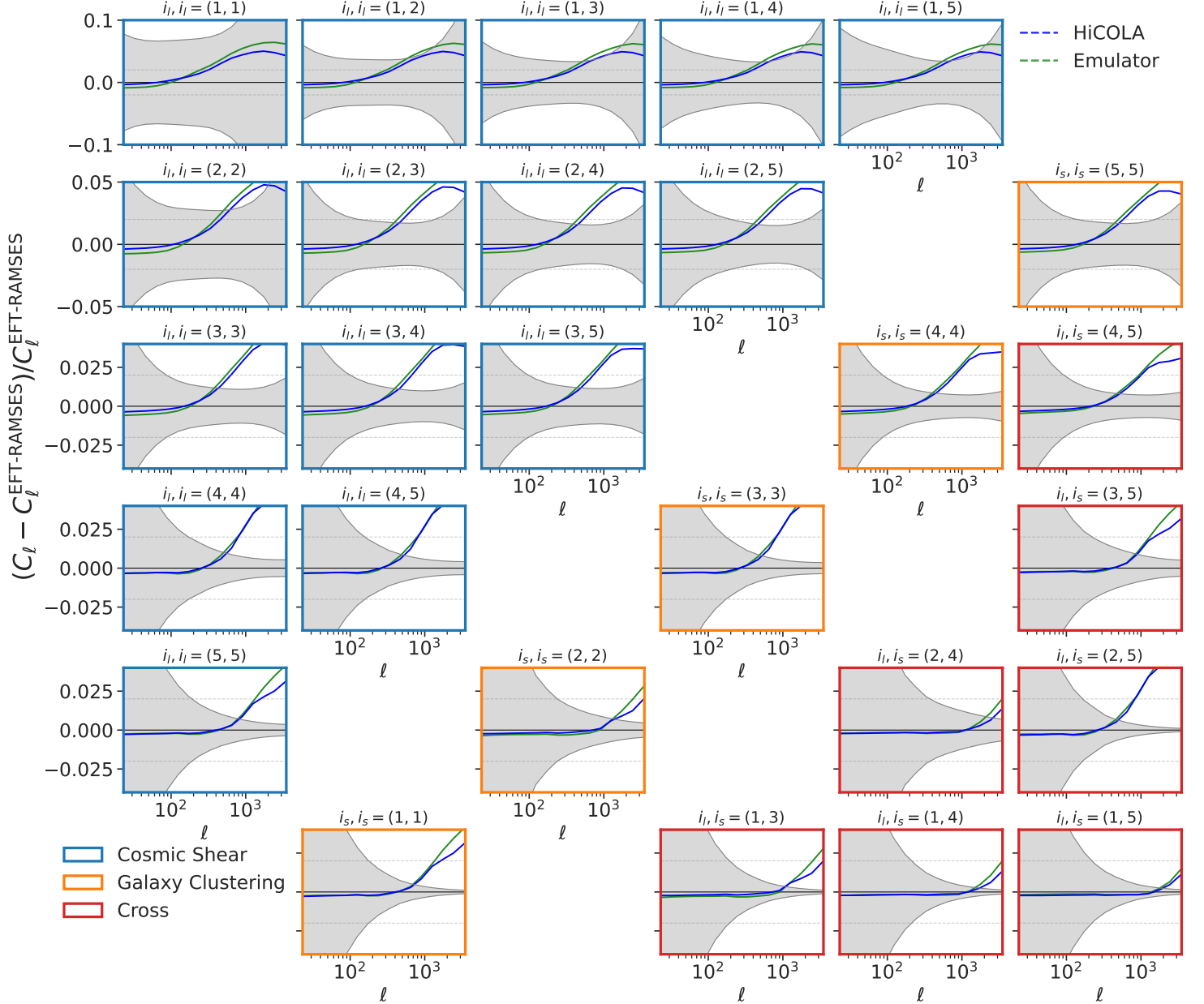


Figure 2: Accuracy of the emulator boost and Hi-COLA boost relative to N-body simulations at the level of the angular power spectra. We display the residual of the Hi-COLA and emulator simulation prediction relative to the EFT-RAMSES prediction (with $f_\phi = 1$), $(C(\ell) - C^{\text{EFT-RAMSES}}(\ell))/C^{\text{EFT-RAMSES}}(\ell)$. The shaded area estimates the (uncorrelated) errors in the C_ℓ values using the DESC SRD LSST Y1 covariance: $\epsilon(\ell_i)/C^{\text{Emu}}(\ell_i) = \sqrt{Cov_{ii}}/C^{\text{Emu}}(\ell_i)$.

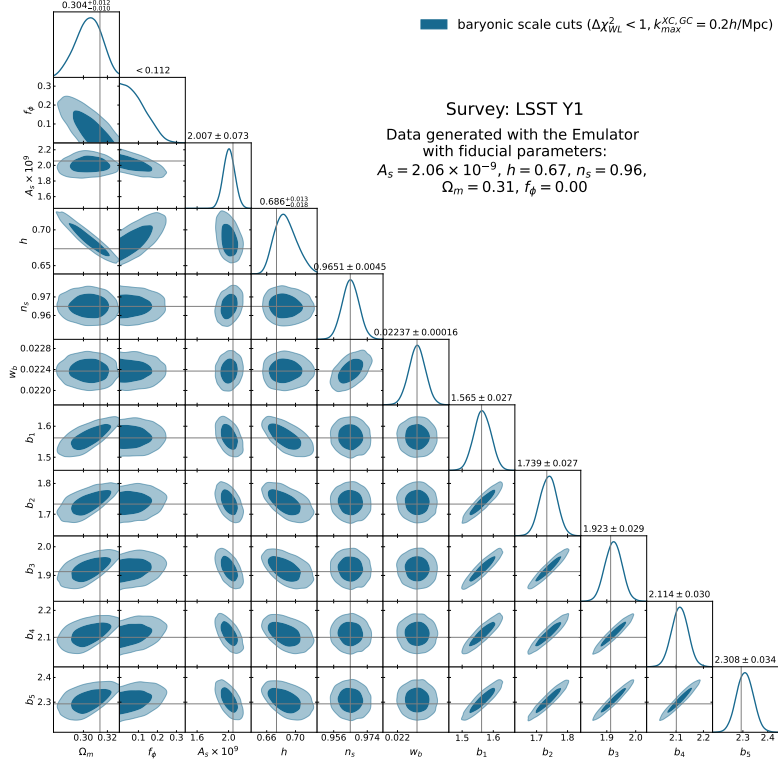


Figure 3: Posterior constraints from an LSST Y1 3×2 pt forecast using mock data generated in GR, analysed with the Cubic Galileon emulator and Planck priors. Unmarginalized posteriors for Figure 8.

B Unmarginalized contour plots

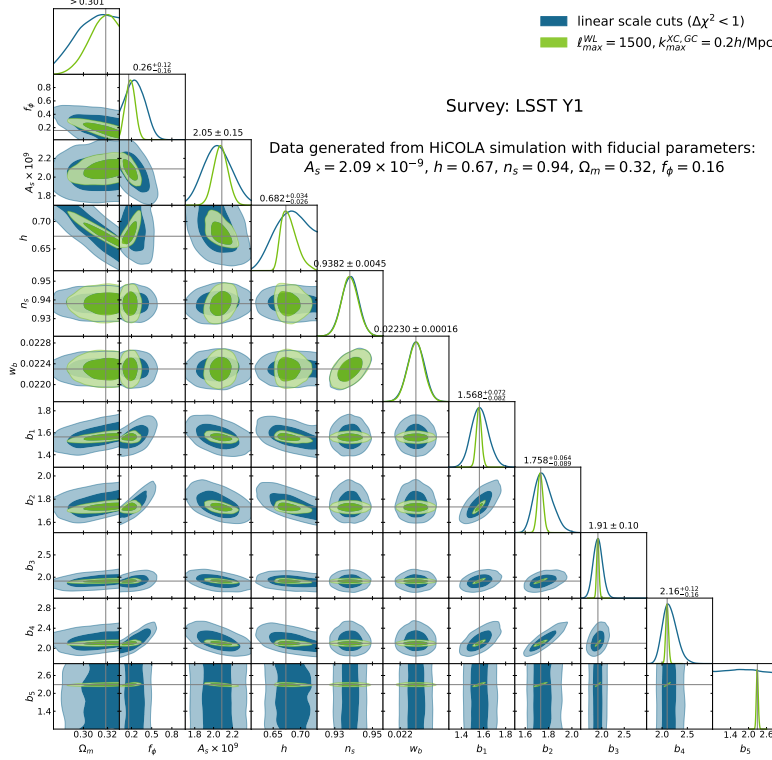


Figure 4: Posterior constraints from an LSST Y1 3×2 pt forecast using mock data generated from Hi-COLA for a validation point located near the fiducial cosmology within the LHS (with $f_\phi \approx 0.16$). The theoretical model uses the Cubic Galileon emulator. Unmarginalized posteriors for Figure 9.

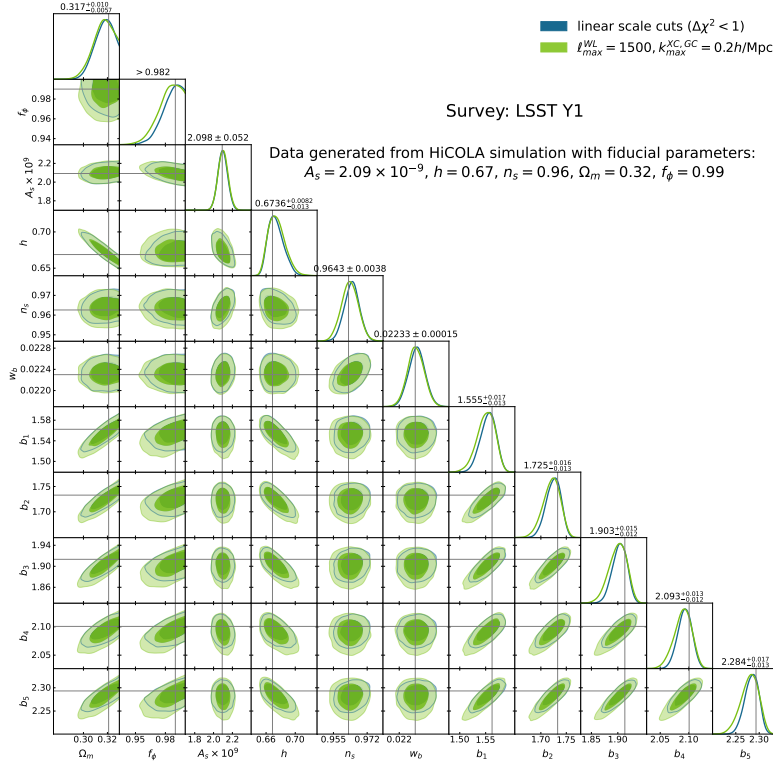


Figure 5: Posterior constraints from an LSST Y1 3×2pt forecast using mock data generated from Hi-COLA for a validation point located near the fiducial cosmology within the LHS (with $f_\phi = 0.99$). The theoretical model uses the Cubic Galileon emulator. Unmarginalized posteriors for Figure 10.

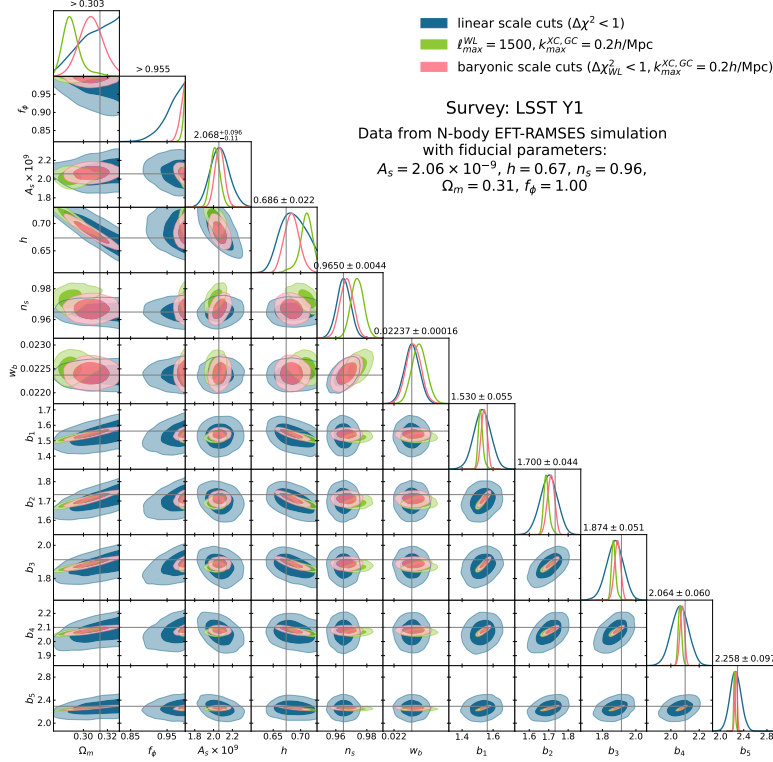


Figure 6: Posterior constraints from an LSST Y1 3x2pt forecast using mock data generated from an EFT-RAMSES N-body simulation near the fiducial cosmology (with $f_\phi = 1$). The theoretical model uses the Cubic Galileon emulator. Unmarginalized posteriors for Figure 12.